Symmetries and Anomalies of Kitaev Spin-S Models: Identifying Symmetry-Enforced Exotic Quantum Matter

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Motivations

Model:

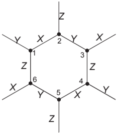
$$H = -\sum_{\langle ij \rangle} J_{\alpha_{ij}} S_i^{\alpha_{ij}} S_j^{\alpha_{ij}}$$

Useful property

$$e^{i\pi S^\mu} e^{i\pi S^\nu} = \epsilon_{\mu\nu\lambda} e^{i\pi S^\lambda} \quad , \quad e^{i\pi S^\mu} e^{i\pi S^\nu} = (-1)^{2S} e^{i\pi S^\nu} e^{i\pi S^\mu}$$

Conserved $\mathbb{Z}_2$ flux:

$$W_p = \prod_{j \in \partial p} e^{i\pi S_j^{\hat{n}_j}}$$



Observations: Even-odd effect (H. Ma, PRL 130, 156701 (2023))

- $S \in \mathbb{Z} + \frac{1}{2}$: gauge charge is **fermionic** and the $\mathbb{Z}_2$ gauge field is **deconfined**.
- $S \in \mathbb{Z}$: gauge charge is **bosonic** and the $\mathbb{Z}_2$ gauge field can be Higgsed.

Questions:

- To what extent is the even-odd effect stable against perturbation?
- Can the results be obtained via a more elementary, direct approach (via spins instead of partons)?

Symmetries

- (Global) $\mathbb{Z}_2^{[1]}$ 1-form symmetry: Generators are two global loops (on a torus)

$$\prod_{j\in\mathcal{L}_v} e^{i\pi S_j^{\hat{n}_j}} \quad , \quad \prod_{j\in\mathcal{L}_h} e^{i\pi S_j^{\hat{n}_j}}$$

- $\mathbb{Z}_2^T$ time reversal symmetry: $S_i^\mu \to -S_i^\mu$
- $\mathbb{Z}_2^{(x)} \times \mathbb{Z}_2^{(y)}$: Generators are

$$\prod_j e^{i\pi S_j^x} \quad , \quad \prod_j e^{i\pi S_j^y}$$

Notice that $e^{i\pi S_j^x} e^{i\pi S_j^y} = e^{i\pi S_j^z}$

Full internal symmetry: $\mathbb{Z}_2^{[1]} \times \mathbb{Z}_2^T \times \mathbb{Z}_2^{(x)} \times \mathbb{Z}_2^{(y)}$

Anomalous 1-Form Symmetry

Anomaly: Presence of obstruction to gauging symmetries. Here we discuss the 1-form symmetry $\mathbb{Z}_2^{[1]}$.

- The action of a generator of $\mathbb{Z}_2^{[1]}$ symmetry

$$S_k^\mu \to \left(\prod_{j\in\mathcal{L}} e^{i\pi S_j^{\hat{n}_j}} \right) S_k^\mu \left(\prod_{j\in\mathcal{L}} e^{i\pi S_j^{\hat{n}_j}} \right)^\dagger = \lambda_k^\mu S_k^\mu$$

  where

$$\lambda_k^\mu = \begin{cases} -1, & k \in \mathcal{L},\ \hat{n}_k \in \mathcal{L} \\ 1, & \text{otherwise.} \end{cases}$$

- Introduce the $\mathbb{Z}_2$ gauge field A_{ij} obey the rule of gauge transformation: $A_{ij} \to \lambda_i^{\alpha_{ij}} A_{ij} \lambda_j^{\alpha_{ij}}$, and then insert them in between matter fields

$$H = -\sum_{\langle ij \rangle} J_{\alpha_{ij}} S_i^{\alpha_{ij}} S_j^{\alpha_{ij}} \to -\sum_{\langle ij \rangle} J_{\alpha_{ij}} S_i^{\alpha_{ij}} A_{ij} S_j^{\alpha_{ij}}$$

- Define the electric fields E_i^μ

$$(E_i^\mu)^2 = \mathbb{1} \quad , \quad E_i^{\alpha_{ij}} A_{ij} E_i^{\alpha_{ij}} = -A_{ij} \quad , \quad [E_i^\mu, E_j^\nu] = 0 \quad , \quad [E_i^\mu, S_j^\nu] = 0$$

Anomalous 1-Form Symmetry

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- Define the Gauss law (generators of gauge symmetries)